

Hanan Abdulwahab Siala

Curriculum Vitae

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Education

- 2023–present **Ph.D. in Computer Science**, Faculty of Natural, Mathematical & Engineering Sciences, King's College London, London, UK
- Thesis: *Model-driven Approaches for Reverse Engineering*
 - Supervisor: **Prof. Kevin Lano**
 - Supervisor: **Prof. Gunel Jahangirova**
- 2004–2007 **M.Sc. in Computer Science**, The Libyan Academy, Tripoli, Libya
- Thesis: *Transformation of Object-Oriented Source Code to UML Models*
 - Supervisor: **Dr. Fauzi Abokhzam**
- 1989–1993 **B.Sc. in Computer Science**, Tripoli University, Tripoli, Libya
- 1987–1989 **General Secondary School Certificate**, Ramadan Ben Turkia Secondary School, Tripoli, Libya

Research Interests

- **Large Language Models (LLMs) for Software Engineering**
My research explores the application of large language models to support software engineering activities, with a focus on automating the extraction, transformation, and understanding of software artifacts. I investigate LLM-based approaches for generating high-level abstractions, recovering design knowledge, and bridging the gap between implementation artifacts and software models.
- **Model-Driven Software Engineering (MDSE)**
My research focuses on leveraging model-driven engineering techniques to automatically derive and maintain software models from existing source code. In particular, I investigate the use of LLMs for extracting UML and OCL representations from Java and Python programs to enhance software documentation, reverse engineering, model recovery, and consistency analysis between implementation and design artifacts.
- **Software Reverse Engineering and Model Recovery**
My research focuses on applying advanced AI techniques to recover high-level software models from existing source code, enabling improved software comprehension, documentation, and maintenance. This research supports the recovery of design knowledge from legacy systems and facilitates the analysis and evolution of complex software applications.

Honors & Awards

- 2023-2027: Scholarship from the Libyan Ministry of Higher Education, to support my Ph.D. studies in London – UK.
- 2014-2015: Scholarship from the Libyan Ministry of Higher Education, to study English for academic purposes in Manchester – UK.

Publications

Journal Papers

- Lano, Kevin and **Siala, Hanan Abdulwahab**, “Extending OCL for semantic representation of software systems,” in *Innovations in Systems and Software Engineering*, vol. 22, no. 1, pp. 12, 2026. <https://link.springer.com/article/10.1007/s11334-026-00639-3>
- **Siala, Hanan Abdulwahab** and Lano, Kevin, “Leveraging LLMs for Abstracting UML/OCL Representations from Java and Python Programs,” in *JSS*, 2026. <https://www.sciencedirect.com/science/article/pii/S016412122600107X>
- **Siala, Hanan Abdulwahab** and Lano, Kevin, “A Comparison of Large Language Models and Model-Driven Reverse Engineering for Reverse Engineering,” in *Frontiers in Computer Science*, 2025. <https://www.frontiersin.org/journals/computer-science/articles/10.3389/fcomp.2025.1516410/full>
- Lano, Kevin and **Siala, Hanan Abdulwahab**, “Using model-driven engineering to automate software language translation,” in *Automated Software Engineering*, vol. 31, no. 20, 2024. <https://link.springer.com/article/10.1007/s10515-024-00419-y>
- **Siala, Hanan Abdulwahab**, Lano, Kevin, Alfraihi, H., “Model-Driven Approaches for Reverse Engineering - A Systematic Literature Review,” in *IEEE Access*, IEEE, 2024. <https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=10509696>
- Abokhzam, Fauzi and **Siala, Hanan Abdulwahab**, “Transformation of Object-Oriented Source Code to UML Models,” in *Libyan Information and Communication Journal*, no. 7, 2007.

Journal Papers (Submitted)

- **Siala, Hanan Abdulwahab** and Lano, Kevin, “Leveraging Large Language Models for Abstracting UML Class Diagrams from Python Programs,” in *JOT*.

Conference Papers

- Lano, Kevin, **Siala, Hanan Abdulwahab**, and Jin, K., “Comparing LLM-based and Model-driven program translation,” in *8th International Conference on Software and System Engineering (ICoSSE)*, Nice, France, 2025. <https://ieeexplore.ieee.org/document/11087925>
- **Siala, Hanan Abdulwahab** and Lano, Kevin, “Using LLMs to Extract UML Class Diagrams from Java and Python Programs: An Empirical Study,” in *Proceedings of the STAF 2025 Workshops: AgileMDE 2025*, Germany, 2025. <https://ceur-ws.org/Vol-4122/paper12.pdf>

- **Siala, Hanan Abdulwahab** and Lano, Kevin, “Using LLMs to Extract OCL Specifications from Java and Python Programs: An Empirical Study,” in *Proceedings of the STAF 2025 Workshops: OCL 2025*, Germany, 2025. <https://ceur-ws.org/Vol-4122/paper2.pdf>
- **Siala, Hanan Abdulwahab** and Lano, Kevin, “Using Large Language Models to Extract UML Class Diagrams from Java Programs,” in *8th International Conference on Software and System Engineering (ICoSSE)*, Nice, France, 2025. <https://ieeexplore.ieee.org/document/11087938/>
- **Siala, Hanan Abdulwahab** and Lano, Kevin, “Towards Using LLMs in the Reverse Engineering of Software Systems to Object Constraint Language,” in *SANER 2025: IEEE International Conference on Software Analysis, Evolution and Reengineering*, IEEE, 2025. <https://conf.researchr.org/home/saner-2025>
- Lano, Kevin and **Siala, Hanan Abdulwahab**, “Using OCL for verified re-engineering,” in *Proceedings of the ACM/IEEE 27th International Conference on Model Driven Engineering Languages and Systems*, Austria, 2024. <https://dl.acm.org/doi/10.1145/3652620.3687824>
- **Siala, Hanan Abdulwahab**, “Enhancing Model-Driven Reverse Engineering Using Machine Learning,” in *46th International Conference on Software Engineering (ICSE 2024)*, Doctoral Symposium, Portugal, 2024. <https://dl.acm.org/doi/10.1145/3639478.3639797>
- Eletri, Amal Ahmed and **Siala, Hanan Abdulwahab**, “Transformation of Requirements in Arabic to UML Diagram,” in *the fourth Scientific Conference of Bright star University of Brega*, Brega, Libya, 2018. <https://fcem.bsu.edu.ly/FCEMBook.aspx>

Presentations

- **Siala, Hanan Abdulwahab**, “Model-driven Approaches for Reverse Engineering,” in *MDENet Annual Symposium 2023*, London, UK, 2023. <https://mdenetwork.com/annual-symposium-2023/>
- **Siala, Hanan Abdulwahab**, “Using Large Language Models to Extract UML Class Diagrams from Java Programs,” in *SSY Group Meeting*, King’s College London, London, UK, Tue 25/03/2025.

Teaching Experience

- 2010-2023 **Lecturer**, Tripoli University, Department of E-commerce and Data Analysis, Tripoli, Libya
- 2008-2010 **Part-time Lecturer**, Tripoli University, Department of E-commerce and Data Analysis, Tripoli, Libya
- 2008-2010 **Part-time Lecturer**, Higher Institute for the Medical Profession, Department of Medical Information Technology, Tripoli, Libya
- 2008-2010 **Part-time Lecturer**, Tripoli University, Department of Computer, Teachers Training College, Tripoli, Libya

Courses Taught

As Lecturer, including lectures and preparation (2008-2023)

- Database Management and Design
- Introduction to Internet and Networks
- Programming Languages
- Fundamentals of Computers
- C++ Programming
- Computer Applications
- Object-oriented Programming (JAVA)
- Web Sites Design

Supervision

- 2010-2023 **Supervisor of four graduate students (projects)**, in the Department of Medical Information Technology and in the Department of E-commerce and Data Analysis, Tripoli University, Libya.

Industrial Experience

Management Experience

- 2009-2010 **Head of Databases and Information Department**, *Information and Documentation Centre, Ministry of Education*, Tripoli, Libya.
- 2008-2009 **Head of the Programming Department**, *Information and Documentation Centre, Ministry of Education*, Tripoli, Libya.

Committee Membership

- 2010 **Member of the Committee on Educational Institutions Data Revision**, *at the national level*, Information and Documentation Centre, Ministry of Education, Tripoli, Libya.
- 2009 **Member of the Follow-up Committee regarding Staff Relocation of the Education Ministry**, *at the national level*, Information and Documentation Centre, Ministry of Education, Tripoli, Libya.
- 2008 **Member of the Human Resources Committee of the Education Ministry**, *at the national level*, Information and Documentation Centre, Ministry of Education, Tripoli, Libya.
- 2002-2003 **Member of the Finance Committee for the Development of the Budget Information System at the Head Office**, Social Security Fund, Tripoli, Libya.
- 1995-1997 **Member of the Pensions committee**, *at the national level*, Social Security Fund, Tripoli, Libya.

Software Engineering Experience

- 2010-2023 **Tripoli University**, Tripoli, Libya
- Designed an application for electronic exams for students.
- 2005-2010 **Information and Documentation Centre, Ministry of Education**, Tripoli, Libya

- Migrated data from an old education system to a new SQL database.
 - Designed and issued statistics and reports for all education levels (elementary, secondary, and higher education).
 - Developed a software system for the management of educational supplies.
 - Developed a Student Information System to supply the statistics and reports needed by the Ministry.
 - Trained the employees of the information and documentation centers at the Ministry of Education on the Student Information System.
 - Developed technical specifications for the systems of the education sector.
 - Contributed to the preparation of annual and quarterly reports for the Information Center at the Ministry of Education.
 - Developed a Human Resources (HR) software system.
- 2003-2006 **National Center for Quality and Informatics**, (*part-time*), Tripoli, Libya
- Developed a Reservation System for the Arab Maghreb Village.
 - Developed a Reservation System for the Tunisian Airline.
- 1994-2005 **Social Security Fund**, Tripoli, Libya
- Developed many software systems for the headquarters of the Libya Social Security Fund and its branches, in support of actuarial studies, pensions, services provided, etc.
 - Developed a registration software system of employers, secured, and collection of security contributions. This system is still in use by the Libya Social Security Fund.
 - Developed a software system to accommodate all retirement files. This system is still in use by the Libya Social Security Fund.

Languages

- **English:** Fluent
- **Arabic:** Native

Hobbies and Interests

- Reading I enjoy exploring new ideas and continuously expanding my knowledge.
- Drawing I like expressing my creativity and improving my artistic skills through drawing.
- Walking I enjoy walking as a way to relax, reflect, and maintain a healthy lifestyle.

References

- **Dr. Kevin Lano**, Professor
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- **Dr. Gunel Jahangirova**, Professor
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